

Nóvo mŕaj $\langle \cdot, \cdot \rangle$ na $\mathbb{R}^2$ jedyŕa $\langle u, v \rangle = u^T A v$. Máme dokázat, že
 existuje báza B taká, že každý x splŕe jŕa $\langle x, x \rangle \geq 0$ a $\langle x, x \rangle = 0$ práke
 standardní sk. souř. $x = 0$.

- Rotations in $\mathbb{R}^2 = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$ for $\alpha \in \mathbb{R}$

Per $\vec{x} = \begin{pmatrix} \cos \alpha \\ \sin \alpha \end{pmatrix}$ e $\vec{y} = \begin{pmatrix} -\sin \alpha \\ \cos \alpha \end{pmatrix}$ gli- ne shadow. di. spinn:

$$\cos \alpha (-\sin \alpha) + \sin \alpha \cos \alpha = \cos \alpha (-\sin \alpha + \sin \alpha) = \underline{\underline{0}} \Leftrightarrow \text{jein Polynom}$$

(usually $\propto$)

- definieren fungsi $f(\alpha) = \left\langle \begin{pmatrix} \cos \alpha \\ \sin \alpha \end{pmatrix}, \begin{pmatrix} -\sin \alpha \\ \cos \alpha \end{pmatrix} \right\rangle$ (d. sein wies rotieren)

for $\alpha = 0$: $f(0) = \left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle = \boxed{0}$

for $\alpha = \frac{\pi}{2}$: $f(\frac{\pi}{2}) = \langle \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \end{pmatrix} \rangle = \langle \begin{pmatrix} 0 \\ 1 \end{pmatrix}, (-1) \begin{pmatrix} 1 \\ 0 \end{pmatrix} \rangle =$

$$\stackrel{\text{d.h. Einsatz}}{=} - \left\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle \stackrel{\text{d.h. symmetrisch}}{=} - \left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle =$$

$$= \boxed{-C} \rightarrow \text{folia pe mână cronometră și obținem } 90^\circ$$

→ de la Bolzano este (de la o jumătate de oraș) într-o semestr na valerabilă

analiza I) mine, ze pahl spajita' funder valgia bodovat s opozicim
samenem, pa se na njekih lodi pona' 0

$\Rightarrow$ since symmetric, $\bar{x}$ $f(x)$ is odd & deriv 0

jez na standardnim st. jeziku, lat - neobavezno